AfyaPlus | Cost-Optimised AI Deployment

EXECUTIVE BRIEF | OCTOBER 2026 | USD SCENARIO ESTIMATES

Decision: Conditional GO

Approve a small, human-reviewed pilot using a managed model service. NO-GO for autonomous clinical decisions or production 10x traffic until the budget, current rates, response-time target, safety checks, and fallback are evidenced. The runnable local service is a deterministic clinic-catalog fallback, not live model inference.

Cost and planned traffic spike

Scenario	Requests/month	Total per 1,000	Monthly estimate
Baseline	100,000	$5.3790	$537.90
Sustained 10x	1,000,000	$0.9790	$979.01

At 10x, the estimate leaves only $20.99 under the proposed $1,000 monthly alert budget. Assumptions: 450 input tokens, 180 output tokens, 2% retries, and OpenRouter-listed rates of $0.15 per million input and $0.60 per million output tokens, checked 2026-10-11. Hosting and staff costs are separate planning allocations, not quotes.

Two measured cost levers

Public-answer cache: 5/8 sample questions served from cache (62.5%). Estimated model-token spend falls from $0.17901 to $0.06713 per 1,000 requests. Answers expire after five minutes. Exact fixture matches do not prove medical correctness; never cache patient-specific triage. **Offline queue:** 7 sample jobs grouped into 2 work groups (71.4% fewer). This is local grouping only; no provider discount or production speed gain is claimed.

Main risk and controls

A traffic or retry spike could exceed budget and slow replies. Send notices at $800 actual spend, $1,000 actual spend, and a $1,000 forecast; identify spend with cost-center, environment, and workload tags; cap simultaneous work and retries; stop repeated failing calls; and send uncertain cases to human review. Budget notices alert people but do not stop spending.

Approval conditions

Keep the interactive response-time target under 2 seconds at the 95th percentile: 95 out of 100 requests must finish within 2 seconds. Before expansion, verify current prices, pass a controlled load test and safety-quality review, stay below the budget limit, and retain the local fallback. No paid cloud, GPU, Redis, or live 10x test was used; Azure/AWS files are configuration examples only.

Recommendation: pilot with these controls; do not approve autonomous clinical use or claim 10x readiness until every condition above is met.